

Carolina Del Pilar Díaz

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Summary

Senior AI Engineer & Applied Statistician (Economist) with 20+ years of experience operating as a Strategic AI/ML Consultant & Forward Deployed Engineer (FDE) for global enterprises. I specialize in architecting Multi-Agent Systems (GenAI, RAG, MCP) that are mathematically robust, preventing model hallucinations in regulated sectors. I have architected and shipped full-stack AI applications, combining the power of Python (n8n, LangGraph) with the rigor of Advanced Statistics, Causal Inference, and MLOps on on-premises, hybrid, and cloud infrastructures across Fintech, AgTech, and Healthcare.

Experience

IT Patagonia (Consultant FDE), Senior AI Engineer

Buenos Aires, Argentina
June 2024 – present

- Multi-Agent Architecture & Cloud Orchestration: Architected a proprietary Multi-Agent Framework and shipped autonomous AI apps for top-tier financial institutions (Banco Galicia, ICBC).
- Designed agnostic architectures that dynamically switch between Google Gemini, OpenAI GPT, and Anthropic Claude via n8n workflows on on-premises, hybrid, and cloud servers.
- Solved critical 'black box' failures by injecting rigorous statistical validation layers into the agent's reasoning loop, ensuring data sovereignty and output reliability.

ITP Patagonia - ICBC, Senior Data Scientist

Buenos Aires, Argentina
June 2024 – July 2025

- Developed predictive models for payroll clients and mutual fund investment using AWS SageMaker, AutoML, Canvas, and Neural Network and XGBoost.
- Built an anomaly detection model using Neural Networks (AENN), Isolation Forest, and SHAP for interpretability using AZURE framework.
- Currently developing a fraud detection model with AWS Fraud Detector, leveraging Decision Trees, Neural Networks, Time Series Analysis, and Custom Fraud Prediction using AutoML.

Nubiral S.A., Senior Data Scientist. COE, Commission of Expert. Generative Models.

Buenos Aires, Argentina
Oct 2023 – May 2024

- Airline solution: Integrated AWS Rekognition, Lambda, S3, LangChain, and OpenAI GPT for Computer Vision and Generative AI, improving object detection accuracy at Aerolineas Argentina.
- Developed a Post-Call Analytics solution using AWS services (Transcribe, Comprehend, Lambda, QuickSight) to analyze customer interactions.
- Implemented Azure Cognitive Search and Form Recognizer for efficient PDF document management at Banco Macro.

Pi Data Strategy & Consulting, Semi-Senior Data Scientist, Statistician, Economist

Córdoba, Argentina
Mar 2021 – Oct 2023

- Creating data regression models, predictive data modelling, and analyzing data mining algorithms. Modelling along creating end to end pipeline to deploy. Azure Technology.
- Achieve EDA, Statistical analysis, regression models, classification models, neuronal network.
- Sales Forecast for georeferenced Dataset. Multinomial predictive models applied to the level of customer satisfaction.

National Institute of Agriculture Technology (INTA), Lead Statistician / Head of Statistical Research Manfredi, Córdoba, Argentina
Jan 2012 – Dec 2021

- Designed and implemented experimental designs for all research projects, ensuring robust statistical methodology across multiple areas.
- Applied neural networks, deep learning, and computer vision for UAV and agricultural machinery, enhancing crop health monitoring.

Education

Postgraduate Master's Degree	National University of Córdoba , Applied Statistics	Córdoba, Argentina Jan 2006 – Dec 2008
Diploma	National University of Córdoba , Data Science Machine Learning and applications	Córdoba, Argentina Jan 2019 – Dec 2020
Bachelor's degree	National University of Córdoba , Economics	Córdoba, Argentina Jan 1993 – Dec 1998

Skills

Core Expertise: Machine Learning, Deep Learning, Generative AI (LLMs, Ollama, Hugging Face), Agentic AI (LangGraph, LangChain, LlamaIndex, AutoGen, Semantic Kernel), Computer Vision, Statistical Modeling, Econometrics

Cloud & Tools: AWS (SageMaker, Lambda, Rekognition, Comprehend), Azure, n8n (Orchestration), Snowflake, Python, R, SAS

Languages: Spanish (Native), English (Intermediate), Portuguese (Basic)

Projects

Country Expansion Program Demographic segmentation. Forecast the Future Demand for actual cities. Anheuser-Busch InBev. Pi consulting & Strategies.	Jan 2022
Logistics prediction of the level of customer satisfaction Health Score for NPS. Anheuser-Busch InBev. Pi consulting & Strategies.	Jan 2022
Genotype by Environmental Predictions GDM. Don Mario. Data Science. Pi consulting & Strategies.	Jan 2021
Digital Government Project of the Ministry of Economy and Energy Data Science, Natural Language Process. Pi consulting & Strategies.	Jan 2021
Text Classifications to improve the management system Pernod Ricard. Data Science. Pi consulting & Strategies.	Jan 2021
PNCYO 1127046 Evaluations networks unpublished of germplasm INTA. Statistical Analysis of Genotypes regarding environments. Publications of the results. INTA.	2013-2019
PNBIO 1131042 Applied Genomics to molecular optimization parametrical and non-parametrical statistical models. INTA.	2013-2019
PNBIO 1131043 Bioinformatics and genomics. Assistance and new models in the evaluation of big base data. INTA.	2013-2019

CORDO 1262206	2013-2019
Transfer and training in the application in new experimental designs. INTA.	
CORDO 1262205	2013-2019
Technology in both extensive and intensive crops under irrigation and rainfed. INTA.	

Publications

The bright side of alfalfa viral diseases: The response of coumestrol phytoestrogen content in alfalfa genotypes	Jan 2020
Pilar Silva, M., Basigalup, D. H., <i>Carolina Del Pilar Díaz</i> (Plant Pathology)	
Segregations Analysis of Polymorphic SSR in three populations of F2 of peanut for charcoal tolerance	Jan 2020
Ritta L., Moreno M.V., Mamaní E., <i>Díaz C.</i> , Etchart V. (Peanut Workday)	
Advanced Mixed Generalized linear models, for effects of two levels of residues of Suico extracts on ruminal fermentation in vitro	Jan 2019
<i>Díaz, C. P.</i> , Scorcione Turcato, M.C, Ortiz Chura A. (XXIII Reunión Científica del GAB)	
The influence of insecticide applications on stored soybeans on the level of residues in the crude oil	Jan 2019
Aguilar Roxana, Nassetta Mirtha, <i>Díaz Carolina</i> , Rojas Dante (7th Congress International Food Science and Technology)	
Nutritional chemical composition of fruits of different carob species from northwestern Córdoba	Jan 2019
Martinez MJ, Ledesma, M, Aguilar R, <i>Díaz C</i> , Nassetta M (VII International Congress of Food Sciences and Technology)	
Assessing INTA's Arachis hypogaea Core Collection for reaction to peanut smut	Jan 2018
Mamaní EMC, Rodríguez AV, Cordes G, <i>Díaz CDP</i> , Moreno MV (Conferencia del Genoma del maní)	
Corn Networks: Analysis of genotype-environment. Application of GGE ShuklaModels	Jan 2018
<i>Díaz Carolina Del Pilar</i> , Ferreyra M.Laura (Horizonte magazine)	
Alfalfa (Medicago sativa L.) coumestrol phytoestrogen content in response to genotypes, cultivar and viral disease	Jan 2018
Silva, M., Chiacchiera, S., Mamaní, E., <i>Díaz, C.</i> (Second World Alfalfa Congress)	
Regional Evaluation of INTA Wheat Varieties	Jan 2018
Cordes D., Candela R., <i>Díaz C.</i> (Rural exhibition of Jesús María)	
Cluster and Yield Evaluation for Corn Varieties	Jan 2018
Chincuini D., Druetta A., <i>Díaz C.</i> , Vega C. (Rural exhibition of Jesús María)	

Phenotyping for Drought Tolerance in Sunflower inbred lines Grandon N., Martin E., Cicconi E., <i>Díaz C. P.</i> (32nd Meeting of Plant Physiology Congress)	Jan 2018
Effect of Chemical and Biological control of acid vine rot Cordes, D.D., Grion, A., <i>Díaz C.</i> (40th Argentine Congress of Horticulture)	Jan 2018
Effect of Lippia turbinata G extracts, at two levels of inclusion, on in vitro ruminal fermentation Scorcione Turcato, M.C., Ortiz Chura A., <i>Díaz, C. D. P.</i> (41° Congreso de la Asoc. Argentina de Producción Animal)	Jan 2018
Deltamethrin residues in crude soybean oil Aguilar, R., <i>Díaz, C.</i> , Strada, J. (Periurban Conferences)	Jan 2017
Analysis of Chlorpyrifos methyl residues in crude soybean oil from treated grains in storage Aguilar R., Nassetta M., <i>Díaz C.</i> (IV Ibero-American Congress of Environmental Health)	Jan 2017
Dissipation dynamics of organophosphate insecticides in crude oil of soybean Aguilar R., <i>Díaz C.</i> , Strada J., Nassetta M. (V International Congress of Food Science and Technology)	Jan 2016
Effects of the alfalfa stunt viral complex on different genotypes estimated with Mixed Linear Models <i>Díaz, C.</i> , Silva, M., Mamaní, E., Basigalup, D. (XXI GAB Scientific Meeting)	Jan 2016
Genetic characterization of cultivated peanut genetic resources using microsatellite markers Pozzi, F, Etchart V, <i>Díaz C</i> , Moreno, M. V (Rev. FCA UNCUYO)	Jan 2014

Certificates and Courses

- Agentic AI on AWS. BeSA- Become a Solution Architect. Candidate. (2026)
- Generative AI Engineer Associate. Candidate. (2026)
- Microsoft Certified: Azure AI Fundamentals. Candidate. (2026)
- AWS Academy Cloud Architecting – Candidate, WIC Training Program (2025)
- AWS Academy Graduate - Cloud Foundations (2024)
- Generative AI Large Language Models & ChatGPT Prompt Engineering for Developers. Deep Learning AI. (2023)
- MLOps Fundamentals CI/CD/CT pipelines with Azure (2022)
- Databricks and Apache Spark for Big Data (2022)
- Machine Learning Pipelines with Azure ML Studio & Azure Synapse SQL Pool (2021)
- Bayesian Statistics: Techniques and Models. University of California (2021)
- AWS Academy Graduate - Machine Learning (2020)
- Diploma in Data Science Machine Learning and applications, National University of Córdoba (2020)
- Machine Learning by Professor Andrew Ng, Stanford University / Coursera (2018)
- Brief Course on Machine Learning & Brief Course on Mixed Models (2016)
- Post-Graduate Course of Bayesian Statistics, National University of Córdoba (2016)